Radiation and chest CT scan examinations: what do we know?

In the past 3 decades, the total number of CT scans performed has grown exponentially. In 2007, > 70 million CT scans were performed in the United States. CT scan studies of the chest comprise a large portion of the CT scans performed today because the technology has transformed the management of common chest diseases, including pulmonary embolism and coronary artery disease. As the number of studies performed yearly increases, a growing fraction of the population is exposed to low-dose ionizing radiation from CT scan. Data extrapolated from atomic bomb survivors and other populations exposed to low-dose ionizing radiation suggest that CT scan-associated radiation may increase an individual's lifetime risk of developing cancer. This finding, however, is not incontrovertible. Because this topic has recently attracted the attention of both the scientific community and the general public, it has become increasingly important for physicians to understand the cancer risk associated with CT scan and be capable of engaging in productive dialogue with patients. This article reviews the current literature on the public health debate surrounding CT scan and cancer risk, quantifies radiation doses associated with specific studies, and describes efforts to reduce population-wide CT scan-associated radiation exposure. CT scan examinations of the chest, including CT scan pulmonary and coronary angiography, high-resolution CT scan, low-dose lung cancer screening, and triple rule-out CT scan, are specifically considered.